

Linear Algebra — Week 02

Answer Key & Misconception Notes

EXAMPLE 1 — Answer, step by step

Step 1. Add coordinatewise: $x + y = (1 + 0, -1 + 3, 2 + 4) = (1, 2, 6)$.

Step 2. Scale every coordinate by 2: $2x = (2 \cdot 1, 2 \cdot (-1), 2 \cdot 2) = (2, -2, 4)$.

Common misconception: Forgetting to scale *every* coordinate by 2 (e.g. only doubling the first entry), or adding x and y by mismatching which coordinate pairs with which.

EXAMPLE 2 — Answer, step by step

Step 1. Identify which F^n each tuple lives in: $x = (1, 2, 3) \in F^3$ (three coordinates), $z = (1, 2) \in F^2$ (two coordinates).

Step 2. Recall the definition on Page 1: addition on F^n is only defined between two elements of the *same* F^n .

Step 3. Since $3 \neq 2$, x and z do not live in the same space, so $x + z$ is **not defined**.

Common misconception: Believing you can “pad” the shorter tuple with a 0 or truncate the longer one to make the addition work — the definition simply doesn’t allow it, there’s no fallback rule.

EXAMPLE 3 — Answer, step by step

Step 1. Fix an arbitrary index i with $1 \leq i \leq n$, and expand the i th coordinate of the left side using the definition of scalar multiplication in F^n : $((c + d)x)_i = (c + d)x_i$.

Step 2. Apply distributivity in the field F itself: $(c + d)x_i = cx_i + dx_i$.

Step 3. Rewrite each term using the definition of scalar multiplication again: $cx_i + dx_i = (cx)_i + (dx)_i$.

Step 4. Recognize the right side by the definition of vector addition in F^n : $(cx)_i + (dx)_i = (cx + dx)_i$.

Step 5. Since $((c + d)x)_i = (cx + dx)_i$ holds for the arbitrary index i , it holds for every $i = 1, \dots, n$, so $(c + d)x = cx + dx$.

Common misconception: Writing out the proof for a specific numeric n (e.g. $n = 3$) instead of a generic index i , or skipping the explicit citation of distributivity *in the field F* as its own step.

EXAMPLE 4 — Answer, step by step

Step 1. Add entrywise: $A + B = \begin{bmatrix} 1+0 & 2+(-1) \\ 3+2 & 4+5 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 5 & 9 \end{bmatrix}$.

Step 2. Scale every entry by 3: $3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$.

Step 3. Check shapes for $A + C$: A is 2×2 , $C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is 2×3 . Since the shapes differ, $A + C$ is **not defined**.

Common misconception: Adding the matching 2×2 block of C to A and dropping the extra column, instead of declaring the whole sum undefined.

EXAMPLE 5 — Answer, step by step

Step 1. Compute cB entrywise with $c = 4$: $(cB)_{11} = 4 \cdot \frac{1}{2} = 2$; $(cB)_{12} = 4 \cdot (-1) = -4$; $(cB)_{21} = 4 \cdot \frac{3}{2} = 6$; $(cB)_{22} = 4 \cdot 0 = 0$.

Step 2. Compare each entry of cB to the corresponding entry of $A = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$: they match in all four positions.

Step 3. By the definition of matrix equality (same shape, and every entry equal), $A = cB$.

Common misconception: Judging the matrices “proportional overall” (e.g. by eyeballing that A “looks like” 4 times B) rather than verifying $A_{ij} = (cB)_{ij}$ entry by entry as the definition requires.

EXAMPLE 6 — Answer, step by step

Step 1. Fix an arbitrary entry position (i, j) and expand the left side using scalar multiplication: $(c(A + B))_{ij} = c(A + B)_{ij}$.

Step 2. Expand $(A + B)_{ij}$ using matrix addition: $c(A + B)_{ij} = c(A_{ij} + B_{ij})$.

Step 3. Distribute c in the field F : $c(A_{ij} + B_{ij}) = cA_{ij} + cB_{ij}$.

Step 4. Recognize each term as a scalar-multiplication entry and recombine using matrix addition: $cA_{ij} + cB_{ij} = (cA)_{ij} + (cB)_{ij} = (cA + cB)_{ij}$.

Step 5. Since this holds for the arbitrary entry (i, j) , it holds for every entry, so $c(A + B) = cA + cB$.

Common misconception: Thinking a generic pair (i, j) only covers “one row and one column” and that rows and columns need to be argued separately, rather than realizing a single arbitrary entry covers the whole matrix at once.

EXAMPLE 7 — Answer, step by step

Step 1. Transpose $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 3 \end{bmatrix}$ (a 3×2 matrix) by swapping rows and columns: $A^T = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 3 \end{bmatrix}$, now 2×3 .

Step 2. Check shapes: $B = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \end{bmatrix}$ is also 2×3 , so $A^T + B$ is defined.

Step 3. Add entrywise: $A^T + B = \begin{bmatrix} 1+1 & 2+2 & 0+1 \\ 0+0 & -1-1 & 3+2 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 1 \\ 0 & -2 & 5 \end{bmatrix}$.

Common misconception: Checking compatibility using A ’s original shape (3×2) instead of A^T ’s new shape (2×3) — forgetting that transposing changes which matrices it can be added to.

EXAMPLE 8 — Answer, step by step

Step 1. Fix an arbitrary entry (i, j) and expand the left side using the definition of transpose: $((A + B)^T)_{ij} = (A + B)_{ji}$.

Step 2. Expand using matrix addition (note both indices are ji , not ij): $(A+B)_{ji} = A_{ji} + B_{ji}$.

Step 3. Recognize each term as a transpose entry: $A_{ji} = (A^T)_{ij}$ and $B_{ji} = (B^T)_{ij}$, so $A_{ji} + B_{ji} = (A^T)_{ij} + (B^T)_{ij}$.

Step 4. Recombine using matrix addition: $(A^T)_{ij} + (B^T)_{ij} = (A^T + B^T)_{ij}$.

Step 5. Since $((A+B)^T)_{ij} = (A^T + B^T)_{ij}$ for the arbitrary entry (i, j) , it holds everywhere, so $(A+B)^T = A^T + B^T$.

Common misconception: Swapping only one index when applying the transpose definition (e.g. writing $(A+B)_{ji}$ as $A_{ij} + B_{ji}$) instead of consistently swapping **both** the row and column index on every term.

EXAMPLE 9 — Answer, step by step

Step 1. *Column picture.* Columns of M : $\vec{c}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\vec{c}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. Form $5\vec{c}_1 - 2\vec{c}_2$.

Step 2. Scale: $5\vec{c}_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$, $-2\vec{c}_2 = \begin{bmatrix} -2 \\ -6 \end{bmatrix}$. Add: $M\vec{v} = \begin{bmatrix} 8 \\ -6 \end{bmatrix}$.

Step 3. *Row picture, as a check.* Rows of M (as column vectors): $\vec{r}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\vec{r}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Dot each with $\vec{v} = (5, -2)$: $\vec{r}_1 \cdot \vec{v} = 2(5) + 1(-2) = 8$; $\vec{r}_2 \cdot \vec{v} = 0(5) + 3(-2) = -6$.

Step 4. Both pictures agree: $M\vec{v} = \begin{bmatrix} 8 \\ -6 \end{bmatrix}$.

Common misconception: In the column picture, adding the raw columns without scaling them by the corresponding entries of $\vec{v}$ first; in the row picture, pairing row and vector entries in the wrong order when taking the dot product.

EXAMPLE 10 — Answer, step by step

Step 1. Recall the rule: for XY to be defined, (columns of X) must equal (rows of Y); the result is (rows of X) $\times$ (columns of Y).

Step 2. AB : A is 3×2 , B is 2×4 . Columns of A (2) = rows of B (2) $\checkmark$ — defined, shape 3×4 .

Step 3. BA : columns of B (4) $\neq$ rows of A (3) — undefined. BC : columns of B (4) $\neq$ rows of C (1) — undefined. CA : columns of C (2) $\neq$ rows of A (3) — undefined. AC : columns of A (2) $\neq$ rows of C (1) — undefined.

Step 4. CB : C is 1×2 , B is 2×4 . Columns of C (2) = rows of B (2) $\checkmark$ — defined, shape 1×4 .

Step 5. Summary: defined are AB (3×4) and CB (1×4); undefined are BA , BC , CA , AC .

Common misconception: Assuming a product is defined just because a “2” appears in both matrices’ shapes, rather than checking that the number of *columns of the first* matches the number of *rows of the second*, in that specific order.

EXAMPLE 11 — Answer, step by step

Step 1. Compute XY entry by entry (row of X dotted with column of Y): $(XY)_{11} = 1(3) + 2(1) = 5$; $(XY)_{12} = 1(0) + 2(2) = 4$; $(XY)_{21} = 0(3) + 1(1) = 1$; $(XY)_{22} = 0(0) + 1(2) = 2$. So $XY = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

Step 2. Compute YX the same way: $(YX)_{11} = 3(1) + 0(0) = 3$; $(YX)_{12} = 3(2) + 0(1) = 6$;

$(YX)_{21} = 1(1) + 2(0) = 1$; $(YX)_{22} = 1(2) + 2(1) = 4$. So $YX = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}$.

Step 3. Compare: $XY \neq YX$ (e.g. the $(1, 1)$ entries are 5 vs. 3), confirming matrix multiplication is not commutative in general.

Common misconception: Multiplying entrywise (Hadamard-style: matching entries in the same position) instead of using the row-dot-column rule — tempting here because X and Y are the same shape.

EXAMPLE 12 — Answer, step by step

Step 1. Add pointwise: $(f + g)(x) = f(x) + g(x) = (x^2 + 1) + 2x = x^2 + 2x + 1$.

Step 2. Scale pointwise, distributing to *every* term: $(3f)(x) = 3f(x) = 3(x^2 + 1) = 3x^2 + 3$.

Common misconception: Forgetting to distribute the scalar to *every* term of $f(x)$, e.g. writing $(3f)(x) = 3x^2 + 1$ instead of $3(x^2 + 1)$.

EXAMPLE 13 — Answer, step by step

Step 1. Recall function equality is pointwise: $f = g$ requires $f(x) = g(x)$ for **every** $x \in F$, not just some.

Step 2. Test a negative input, $x = -1$: $f(-1) = -1$, but $g(-1) = |-1| = 1$.

Step 3. Since $f(-1) \neq g(-1)$, the two functions disagree at at least one input, so $f \neq g$.

Common misconception: Concluding $f = g$ because the two functions agree for all $x \geq 0$ — forgetting that function equality (pointwise) requires agreement at **every** input in F , not just most of them.

EXAMPLE 14 — Answer, step by step

Step 1. Fix an arbitrary input $x \in F$ and expand the left side using scalar multiplication of functions: $(c(f + g))(x) = c((f + g)(x))$.

Step 2. Expand $(f + g)(x)$ using function addition: $c((f + g)(x)) = c(f(x) + g(x))$.

Step 3. Distribute c using the field axioms: $c(f(x) + g(x)) = cf(x) + cg(x)$.

Step 4. Recognize each term as a scalar-multiplication value and recombine using function addition: $cf(x) + cg(x) = (cf)(x) + (cg)(x) = (cf + cg)(x)$.

Step 5. Since $(c(f + g))(x) = (cf + cg)(x)$ for the arbitrary input x , the two functions agree everywhere, so $c(f + g) = cf + cg$.

Common misconception: Treating $c(f + g)$ as an algebra shortcut and skipping the definitional chain (function addition, then scalar multiplication, then distributivity in F) instead of justifying each equality as its own step.

EXAMPLE 15 — Answer, step by step

Step 1. $A - B$ means $A + (-1)B$; subtract entrywise: $A - B = \begin{bmatrix} 2 - (-1) & 1 - 2 \\ -3 - 0 & 4 - 5 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -3 & -1 \end{bmatrix}$.

Step 2. Scale B by 2: $2B = \begin{bmatrix} 2(-1) & 2(2) \\ 2(0) & 2(5) \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 0 & 10 \end{bmatrix}$.

Common misconception: Subtracting the matrices “column-major” instead of entry-by-

entry in place (e.g. swapping which entries line up), or forgetting that subtraction scales B by -1 before adding, which matters if a sign is mishandled.

EXAMPLE 16 — Answer, step by step

Step 1. Compute MN (row of M dotted with column of N), $M = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $N = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$:
 $(MN)_{11} = 1(2) + (-2)(-1) = 2 + 2 = 4$; $(MN)_{12} = 1(1) + (-2)(4) = 1 - 8 = -7$;
 $(MN)_{21} = 3(2) + 0(-1) = 6$; $(MN)_{22} = 3(1) + 0(4) = 3$. So $MN = \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$.

Step 2. Compute NM : $(NM)_{11} = 2(1) + 1(3) = 5$; $(NM)_{12} = 2(-2) + 1(0) = -4$;
 $(NM)_{21} = -1(1) + 4(3) = 11$; $(NM)_{22} = -1(-2) + 4(0) = 2$. So $NM = \begin{bmatrix} 5 & -4 \\ 11 & 2 \end{bmatrix}$.

Step 3. Compare: every entry of MN differs from the corresponding entry of NM , so $MN \neq NM$.

Common misconception: Reusing the row-dot-column pairing from MN without recomputing it for NM — the rows and columns being dotted genuinely swap roles, so NM must be computed from scratch, not guessed from MN .

EXAMPLE 17 — Answer, step by step

Step 1. Compute $(f \circ g)(x) = f(g(x))$: substitute $g(x) = x + 1$ into $f(t) = t^2$, giving $f(g(x)) = (x + 1)^2 = x^2 + 2x + 1$.

Step 2. Compute $(g \circ f)(x) = g(f(x))$: substitute $f(x) = x^2$ into $g(t) = t + 1$, giving $g(f(x)) = x^2 + 1$.

Step 3. Compare the two functions: $f \circ g$ has value $x^2 + 2x + 1$ while $g \circ f$ has value $x^2 + 1$ — these agree only when $2x = 0$, i.e. only at $x = 0$, not for every x .

Step 4. Since function equality requires agreement at **every** input, and e.g. at $x = 1$ we get $(f \circ g)(1) = 4 \neq 2 = (g \circ f)(1)$, we conclude $f \circ g \neq g \circ f$.

Common misconception: Assuming composition of functions is commutative the way multiplication of numbers is — it is not, and checking a single input where they disagree is enough to disprove equality, by the definition of function equality on Page 1.